

Budget Optimization

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Mode: Full Pipeline

1. Problem Formulation

Problem Type:	Find
Domain:	Integer Linear Programming
Named Problem:	Knapsack Problem

Universe

- See problem description

Variables

Symbol	Meaning	Type / Range
x_{ij}	assignment decision	$\{0, 1\}$
n	number of binary variables = 10	$\mathbb{Z}^+$

Structure

A startup has a \$500K budget and 10 candidate projects. Each project has a fixed cost and an expected ROI score. Select the subset of projects that maximizes total ROI without exceeding the budget.

Real-World Mapping

Real-World Concept	Mathematical Object
total score/cost	objective function value

Constraints

(1) 0/1 Knapsack as ILP	classic combinatorial optimization with binary variables
(2) Budget-constrained selection	real-world resource allocation under constraints
(3) ROI-driven prioritization	scoring and ranking candidate investments
(4) Sensitivity analysis	effect of budget changes on optimal selection

Objective

$$\max \sum_{i,j} c_{ij} x_{ij}$$

Approach

Suggested approach 0/1 Knapsack ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: 0/1 Knapsack ILP

2. Solution

Answer:	Solution computed
Objective Value:	50.0
Optimal:	Yes
Feasible:	Yes
Algorithm:	0/1 Knapsack ILP (PuLP/CBC, Branch & Bound)
Time:	0.0075s
Certificate:	Branch-and-bound solved to optimality.

Solution Details

Objective value: $z^* = 50.0$

Method: 0/1 Knapsack formulated as Integer Linear Programming via PuLP/CBC.

- 10 binary variables (one per project: selected or not)
- 1 budget constraint (total cost $\leq$ \$500K)
- Objective: maximize sum of ROI scores for selected projects

Verification

✓ Feasibility	All constraints satisfied
✓ Optimality	0/1 Knapsack ILP (PuLP/CBC, Branch & Bound)

3. Interpretation

The Question

A startup has a \$500K budget and 10 candidate projects.

The Answer

The optimal solution has objective value $z^* = 50.0$.

What This Means

- Optimal project selection maximizing total ROI score under \$500K budget
- Budget utilization percentage
- Per-project selection table
- Independent verification of budget and selection constraints

0/1 Knapsack formulated as Integer Linear Programming via PuLP/CBC.

- 10 binary variables (one per project: selected or not)
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- Objective: maximize sum of ROI scores for selected projects

Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution